

Use case flow:

Reserve Charging Slot

Actor: EV driver

Main Success Scenario

1. Driver selects “Reserve Charging Slot” and searches for stations within a chosen radius; system shows compatible stations sorted by distance.
2. Driver selects a station and views its available charger slots.
3. Driver selects a preferred connector type and time slot(start time+ duration).
4. System validates the driver’s vehicle connector type against hold on the slot.
5. System checks socket availability and places a temporary hold on the slot.
6. System displays the estimated dynamic charging cost for that slot.
7. Driver confirms the reservation and proceeds to payment.
8. System sends the payment request to the Payment Gateway.
9. Payment is authorized and the system confirms the reservation.
10. System sends a confirmation code/QR to the driver, and the use case ends successfully.

Alternate Flow- 8a. Payment not successful

8a1. System displays a “payment failed” message.

8a2.Driver is prompted to retry with the same or a different payment method.

8a3. If payment fails again after 2 attempts, the temporary slot hold is released, the reservation is cancelled, and the driver is notified.